

India.Runs Ideathon Track 1 Submission

Copy and paste this directly into your `[IDEATHON] Track 1 Submission Template` slides! Since this hackathon is sponsored by **Redrob**, we have strategically named Annapurna as a new B2B extension of the Redrob AI ecosystem.

Important Links to Include in your Submission:

- * **Live Prototype:** <https://annapurna-gemini-apac.vercel.app/>
- * **GitHub Repo:** <https://github.com/sumitsaraswat362/Annapurna-Gemini-APAC>
- * **Demo Video:** *(Paste your public YouTube link here!)*

Slide 4: Title Slide

* **Team Name:** *(Enter your team name)

* **Team Members:** *(Enter your names)

* **Problem Statement:** Up to 40% of food in emerging markets is wasted during transit due to broken logistics, unpredictable delays, and cold-chain failures. Current telematics are purely reactive, leaving fleet managers unable to mitigate financial losses when hardware fails.

Slide 5: Problem Definition

* **What problem are you solving?** We are solving the catastrophic economic and environmental fallout of cold-chain logistics failures. When a refrigerated truck breaks down, the cargo rots, resulting in a total loss.

* **Who experiences this problem?** B2B Logistics providers, fleet managers, farmers, and wholesale food buyers.

* **Why is the current approach insufficient?** Current logistics dashboards only provide reactive alerts (e.g., "Your truck is warm"). They do not offer proactive, automated economic solutions (like finding a buyer before the food spoils).

Slide 6: Opportunity & Vision

* **Why is this an important opportunity?** Food waste in transit is a multi-billion dollar crisis in India. Solving it recovers massive economic value and prevents millions of tons of methane emissions.

* **What future state are you enabling?** A future where supply chains are entirely self-healing. When hardware fails, the AI-native workflow automatically mitigates the loss by instantly rerouting distressed assets to local B2B buyers without human intervention.

Slide 7: Solution Overview

* **What is your proposed solution?** Annapurna: A predictive AI logistics platform and Emergency Wholesaler Marketplace.

* **What makes it AI-native rather than AI-assisted?** Instead of humans reading telemetry to make decisions, our Gemini-powered engine autonomously evaluates live data (temperature, ethylene, ETA), calculates spoilage windows, and triggers automated SOS liquidation protocols without human input.

* **Which existing Redrob capability does it build upon?** It extends Redrob's AI-native workflow and B2B recommendation engine, adapting its core intelligent matching algorithms to the physical supply chain and logistics sector.

Slide 8: User Journey / Workflow Diagram

* **How does a user interact?** Fleet managers monitor a live dashboard. If an anomaly occurs, the system auto-notifies local Wholesalers, who interact via a

live bidding interface.

- * ****How does info flow?**** Truck Telemetry -> AI Engine -> Live WebSockets -> Wholesaler Marketplace.
- * ****Integration with Redrob:**** The Wholesaler Bidding system integrates directly as a new "Redrob Supply Chain" vertical.
- * ***(Mandatory Visual: Paste the "Process_Flow_Diagram.png" I generated for you yesterday here!)***

Slide 9: AI Logic & Decision Flow

- * ****Where does AI intervene?**** At the edge of hardware failure. As soon as telemetry drifts from safe parameters, the AI intercepts the data stream.
- * ****How are decisions made?**** The AI cross-references current temperature against the max-safe temp and the ETA. If Spoilage Time < ETA, it triggers liquidation.
- * ****How do agents interact?**** The Telemetry Agent streams data to the Gemini Decision Agent, which triggers the Marketplace Matching Agent.
- * ***(Mandatory Visual: You can use a screenshot of the "AI Decision Card" from your Vercel app here)***

Slide 10: System Architecture

- * ****What components make up the system?**** Next.js Frontend, Supabase (PostgreSQL) Real-Time Database, Google Gemini 1.5 Flash API.
- * ****How do services interact?**** Next.js streams simulated IoT data to Supabase. Supabase triggers edge functions to Gemini, which returns structured JSON routing decisions broadcasted via WebSockets.
- * ****Which Redrob systems are leveraged?**** Redrob's intelligent user experience (UX) paradigms and real-time B2B matching workflows.
- * ***(Mandatory Visual: Paste the "Architecture_Diagram.png" I generated for you yesterday here!)***

Slide 11: Data, Context & Intelligence Layer

- * ****What data powers the solution?**** Live IoT truck telemetry (Ambient temperature, humidity, ethylene levels, GPS coordinates).
- * ****How is context utilized?**** Context (traffic delays, weather) is injected into the prompt so Gemini understands the exact physical state of the cargo.
- * ****How does existing Redrob context improve the experience?**** By utilizing Redrob's existing user trust network and verified B2B profiles, we ensure that the wholesalers bidding on distressed cargo are legitimate and capable of instant payment.

Slide 12: Scalability & Technical Feasibility

- * ****How would this be implemented?**** The prototype is already fully built and deployed on Vercel. Enterprise implementation requires API integration with existing fleet telematics (like Samsara or Geotab).
- * ****How does the system scale?**** Built on Supabase (Postgres) and Edge Functions, it horizontally scales to handle millions of concurrent telemetry pings.
- * ****What technical challenges exist?**** Ensuring low-latency mobile connectivity for trucks in rural Indian areas. We mitigate this by edge-caching decisions when offline.

Slide 13: Redrob Ecosystem Integration

- * ****Which Redrob capabilities are leveraged?**** Redrob's B2B discovery and inte

elligent recommendation workflows.

- * **What new capability does it introduce?** "Redrob Logistics" – bridging the gap between digital workflows and physical supply chain recovery.
- * **How does it strengthen the ecosystem?** It brings massive physical enterprise networks (fleets, warehouses, wholesalers) into the Redrob ecosystem.
- * **What additional opportunities become possible?** Using Redrob's core hiring platform to recruit drivers and logistics managers based on the fleet performance data Annapurna collects.

Slide 14: Impact & Success Metrics

- * **Measurable outcomes:** 30% reduction in total transit food loss. 15% recovery of sunk costs for fleet operators during breakdowns.
- * **How will success be tracked?** Total tonnage of food successfully liquidated via the Emergency Marketplace vs. total tonnage spoiled.
- * **What value is created?** Fleet operators recover money from broken trucks. Wholesalers get discounted produce. Redrob captures a massive new B2B market vertical.

Slide 15: Future Roadmap

- * **How could this evolve over 2-3 years?** Expansion from refrigerated food to all perishable cold-chains, including pharmaceuticals (vaccines) and volatile chemicals.
- * **What future capabilities can be unlocked?** Fully autonomous self-driving trucks that automatically reroute themselves based on Annapurna's marketplace bids.
- * **What broader vision does this support?** A zero-waste global supply chain powered entirely by proactive, AI-native decision engines.